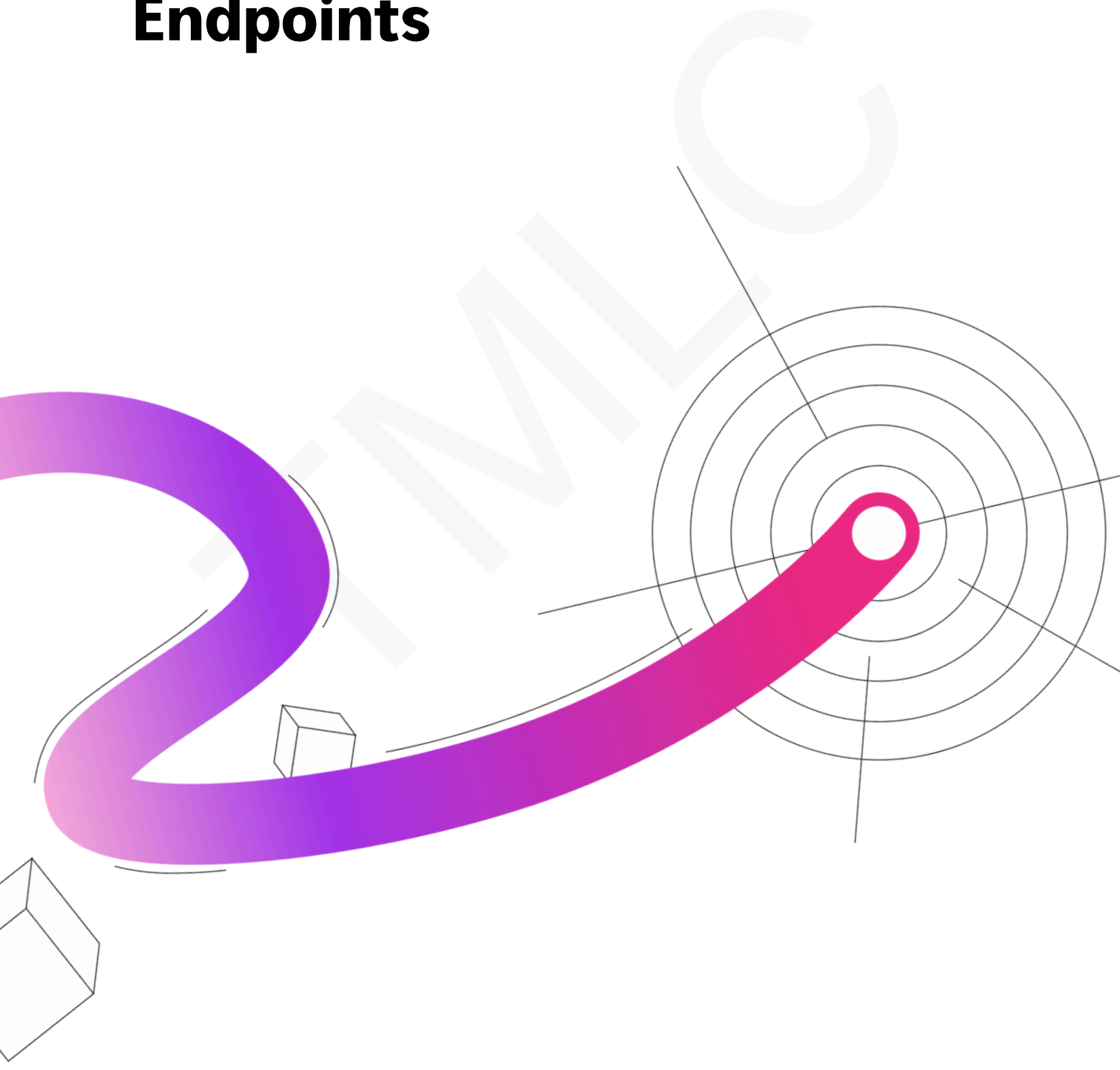


Hugging Face Inference Endpoints



Deploying AI models can be a complex and costly endeavour. **Hugging Face Inference Endpoints** simplifies this process by allowing users to easily deploy Transformers, Diffusers, or any other models on a **fully managed infrastructure**. With a **secure, compliant, and cost-effective solution**, Inference Endpoints empower businesses and developers to scale their AI applications effortlessly.

Accessing 🧐 Inference Endpoints

To access the **Inference Endpoints** web application, you or your organization need to add a **valid payment method** to your Hugging Face account. You can verify your **billing details** [here](#).

Pricing Plans

1. **Pay-as-you-go**: Hourly compute pricing
2. **Enterprise Plan**: Custom pricing for businesses requiring **dedicated support, SLAs, and uptime guarantees**.

Once you've added a payment method, you can start deploying models instantly! 🚀

Creating an Inference Endpoint

Once logged in, you will be directed to the **Endpoint Creation** page.

Access the link at: <https://endpoints.huggingface.co/<USERNAME>/new>

Replace **<USERNAME>** with your account username.

Here's how to deploy a model:

Steps to Deploy:

1. **Enter Repository ID & Endpoint Name** – Provide the **Hugging Face repository ID** and specify your **desired endpoint name**.

 **Create a new Endpoint**

[Explore our Inference Catalog](#) to deploy popular models on optimized configuration.

Model Repository 

Endpoint Name 

microsoft/phi-4 

phi-4

 More options

2. **Select Cloud Provider & Instance Type** – Choose the **provider**, **region**, and **compute instance** based on your needs.

 **Hardware Configuration** 

[Contact us](#) if you'd like to request a custom solution or instance type.

 Amazon Web Services

 Microsoft Azure

 Google Cloud Platform

CPU GPU INF2

 N. Virginia us-east-1 

Nvidia T4
1 GPU · 16 GB
3 vCPUs · 15 GB
\$ 0.5/h

Nvidia L4
1 GPU · 24 GB
7 vCPUs · 30 GB
\$ 0.8/h

Nvidia A10G
1 GPU · 24 GB
6 vCPUs · 30 GB
\$ 1/h

Nvidia L40S
1 GPU · 48 GB
7 vCPUs · 30 GB
\$ 1.8/h

Nvidia T4
4 GPUs · 64 GB
46 vCPUs · 192 GB
\$ 3/h

Nvidia L4
4 GPUs · 96 GB
47 vCPUs · 185 GB
\$ 3.8/h

Nvidia A100
1 GPU · 80 GB
11 vCPUs · 145 GB
\$ 4/h

Nvidia A10G
4 GPUs · 96 GB
46 vCPUs · 186 GB
\$ 5/h

3. Apply Automatic Scale-to-Zero – Enable or disable auto-scaling for cost optimization.

 Autoscaling 

Automatic Scale-to-Zero 

Number of replicas 

After 15 minutes with no activity 

Min

Max

 More options

4. Define Security Level – Configure authentication and access permissions.

 Security Level 

☒ Protected

☐ Public

☐ Private

A Protected Endpoint is available from the Internet, secured with TLS/SSL and requires a valid [Hugging Face Token](#) for Authentication.

5. Customize Advanced Configurations – Adjust replica autoscaling, task type, framework, and container type.

6. Create the Endpoint – Click Create Endpoint, review the cost estimate, and launch it.

\$1.80 / h
per running replica

 Create with cURL

Create Endpoint

The create endpoint button will work upon adding a valid payment information to the billing section.

7. Wait for Deployment – Initialization time varies between 1 to 5 minutes.

Sending Requests to Inference Endpoints

Once your endpoint is live, you can send requests using:

- **UI Inference Widget** (for quick testing)
- **cURL** (Command-line requests)
- **@huggingface/inference Library** (Python & JavaScript)

Example: Using cURL to send a request

```
curl https://your-endpoint-url \  
-X POST \  
-d '{"inputs": "Deploying my first endpoint was an amazing \  
experience."}' \  
-H "Authorization: Bearer <Token>"
```

Updating Your Endpoint

You can **modify an active endpoint's** configuration via the **Settings tab**:

- **Instance Type**: Upgrade/downgrade between CPU/GPU instances.
- **Autoscaling**: Adjust **minimum and maximum replicas**.
- **Task Type**: Modify the **pipeline type** for different AI workloads.
- **Model Revision**: Update to a **newer version** of the deployed model.

If your endpoint is **failing**, you must create a **new one**.

Autoscaling: Dynamic Scaling for Cost Efficiency

Hugging Face **Inference Endpoints** support autoscaling based on usage patterns.

Scaling Criteria:

- **CPU Instances**: A new replica is added when **CPU utilization reaches 80%**.

- **GPU Instances:** A new replica is added when **GPU utilization exceeds 80%** over 1 minute.
- **Scaling Down:** Happens **every 2 minutes** when utilization drops below the threshold.

Scaling to Zero: Cost Optimization

When an endpoint **remains idle for 15+ minutes**, it **automatically scales down to zero replicas** to reduce costs. However, cold starts may introduce slight delays when requests resume.

Supported Models

Inference Endpoints support multiple ML frameworks, including:

- **Transformers** (NLP tasks like Text Classification, Summarization, etc.)
- **Sentence-Transformers** (Embeddings, Semantic Search)
- **Diffusers** (Text-to-Image generation)

Conclusion

Hugging Face **Inference Endpoints** make AI deployment seamless, scalable, and cost-efficient. Whether you're a **researcher, developer, or enterprise**, this solution offers **fully managed infrastructure** to bring your AI models to production with minimal effort.